Waseem Raza, PhD (Electrical and Computer Engineering)

Electrical & Computer Engineering Technology Educator | Applied Labs and Telecommunications

Computer Programming | Communication Systems | Data Networking | Wireless Engineering

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LinkedIn | GitHub | Google Scholar | Coursera Courses | Portfolio

Professional Profile

PhD in Electrical and Computer Engineering (University of Oklahoma, 2025), with previous undergraduate Electrical Engineering Technology teaching at the University of Lahore in Computer Programming, Communication Systems, and Wireless Communication. Experienced supporting communications laboratories at UET Taxila, mentoring engineering students at COMSATS, and connecting theory with practical programming, signal and network analysis, and technical problem-solving. Industry research and engineering with AT&T Labs, Futurewei, and AT&T provide applied telecommunications context.

Education

PhD, Electrical and Computer Engineering *09/2019 - 07/2025*
University of Oklahoma, Norman, Oklahoma, USA

MSc, Telecommunication Engineering *08/2014 - 08/2016*
University of Engineering and Technology, Taxila, Pakistan

BSc, Telecommunication Engineering *11/2010 - 06/2014*
University of Engineering and Technology, Taxila, Pakistan

Teaching & Applied Instruction

University Teaching in Engineering Technology

Lecturer, Department of Technology **12/2017 - 09/2019**
University of Lahore, Pakistan *Lahore, Pakistan*

- Taught programming, communication systems, and wireless communication with practical examples.
- Advised senior-design students on projects, academic plans, and early careers.
- Aligned course materials with learning outcomes and accreditation standards.
- Supported accreditation, exams, technical workshops, and student activities.

Teaching areas: Computer Programming (previously taught); Communication Systems (previously taught); Wireless Communication (previously taught); Communication Systems Laboratories (instructional support); Computer Applications and Data Networking (technical preparation); Signal Processing and Applied Experiments (technical preparation)

Student mentoring and research development

- Advised undergraduate Electrical Engineering Technology projects in programming and communications applications.
- Supported student communication-systems experiments, practical problem solving, technical reports and evaluation.
- Mentored undergraduate and graduate research students in simulations, experiment design, documentation and presentations.

Technical Skills & Engineering Tools

Programming Languages: Python, C/C++, MATLAB, SQL; programming fundamentals, scripting, debugging, computational problem-solving, and numerical computing

Engineering Simulation & Data Tools: MATLAB, Jupyter Notebook, NumPy, SciPy, Pandas, Matplotlib, and Plotly; simulation, signal and network measurement analysis, data visualization, and reproducible experiments

Electrical & Wireless Technology: Analog and digital communication fundamentals, RF and wireless systems, LTE/4G, 5G-NR, signal propagation, network measurements, data communications and performance analysis

AI/ML Software: scikit-learn, PyTorch, TensorFlow, Keras, and XGBoost; classification, regression, anomaly detection, and engineering data analysis

Software Development & Cloud: Git/GitHub, Linux, VS Code, REST APIs, Docker, GitHub Actions and CI/CD; experience with cloud-based data analytics and Python workflows

Teaching, Laboratories & Assessment: Undergraduate Electrical Engineering Technology instruction; communications experiments, lab reports, worked examples, programming exercises, student project mentoring, and learning-outcome assessment

University and Industry Engineering Experience

Research and Laboratory Support

Graduate Research Assistant **09/2019 - 07/2025**
AI4Networks, University of Oklahoma *Tulsa, OK*

- Built 3GPP-compliant Python simulation tools and an AI-enabled 5G research testbed.
- Built GAN, transfer-learning, autoencoder, and reinforcement-learning models for wireless data.
- Led 100+ experiments testing wireless ML under domain shift and limited labeled data.
- Developed self-healing and outage-management methods, improving resilience by 30%.
- Published wireless AI research and mentored junior researchers in experimental design.

Research Associate, ComSens Lab **06/2016 - 12/2017**
COMSATS University Islamabad, Pakistan *Islamabad, Pakistan*

- Modeled wireless communication systems through simulation, analysis, and technical reports.
- Mentored Bachelor's and Master's theses in experiment design and research presentation.
- Reviewed student reports and manuscripts for rigor and publication readiness.
- Supported proposals, literature reviews, lab seminars, and research coordination.

Full-Time Research Scholar, Communication Systems Lab **08/2014 - 02/2016**
University of Engineering and Technology, Taxila, Pakistan *Taxila, Pakistan*

- Researched wireless communication systems while completing an MSc in Telecommunications.
- Supported communications courses, labs, assignments, grading, and examinations.
- Led wireless communication labs and prepared technical manuals and experiments.
- Developed course and student-project materials for practical wireless engineering.

Industry Research and Engineering

Machine Learning Engineer **02/2025 - Present**
AT&T, contractor via Innovatix Inc. *Bedminster, NJ*

- Applied domain-adaptive ML to telecom service assurance and model validation.
- Designed cross-domain evaluation and error-analysis methods for operational ML.
- Worked with engineering and public-safety teams on reliable network assurance.

Data Scientist and AI/ML Engineer **09/2024 - 12/2024**
Impac Exploration Services *Weatherford, OK*

- Built LLM-enabled analytics to interpret geochemical well data for decision support.
- Built Python pipelines in Azure Databricks, Functions, and Synapse Analytics.
- Automated exploration-data ETL, cutting preparation time by 40%.
- Used data-validation and quality controls to improve output reliability by 20%.
- Turned geologists' technical needs into structured datasets and useful analytics.

Advanced-Degrees Intern **06/2024 - 08/2024**
AT&T Labs *Austin, TX*

- Analyzed 500K+ LTE/5G-NR logs with ML, reducing diagnostic time by 20%.
- Improved 5G NR-NTN cell selection by 15% through data-driven validation.
- Presented operational data insights to next-generation wireless engineering teams.

Advanced Wireless Technology Development Intern **09/2022 - 01/2023**
Futurewei Technologies *Schaumburg, IL*

- Improved wireless CSI compression by 25% with transfer learning and autoencoders.
- Ran 50+ experiments on wireless ML robustness under changing data conditions.
- Translated research results into recommendations for future wireless networks.

Advanced-Degrees Intern **06/2022 - 08/2022**
AT&T Labs *Bedminster, NJ*

- Improved 5G standalone user-timing precision by 20% with ML and statistics.
- Analyzed user traces with SQL and presented results to engineering leadership.
- Turned operational traces into indicators for network diagnosis and optimization.

Selected Applied Engineering Projects

SyntheticNET: 3GPP-Compliant Network Simulation *Research Project | 2020-2024*

Developed a Python-based, 3GPP-aligned simulation environment for realistic mobility, propagation, and cellular-network experimentation when measured data is sparse or costly. Supported controlled data generation and repeatable studies in propagation modeling, network optimization, digital twins, and resilient wireless AI.

Tools & Methods: Python, 3GPP, Wireless Simulation, Mobility Modeling, Propagation Modeling, Synthetic Data, Experiment Design, Digital Twins, Network Analytics

Propagation Models & Digital Twins *Research Project | 2024*

Developed robust data-driven radio-propagation models for wireless digital twins using domain-informed generative learning and scarce measured data. Combined 3GPP-informed pretraining, GAN-based augmentation, domain adaptation, and statistical validation to improve resilience under distribution shift.

Tools & Methods: Propagation Modeling, Digital Twins, GANs, Domain Adaptation, 3GPP Models, Synthetic Data, Statistical Validation, Wireless AI, Distribution Shift

TurboRAN Experimental Platform *Research Infrastructure | 2020-2024*

Built and used a multi-band, multi-tier experimental platform for evaluating AI-enabled radio-access-network methods under controlled network conditions. Supported repeatable measurement collection, testbed experimentation, RAN optimization studies, and validation of emerging wireless-network algorithms.

Tools & Methods: 5G Testbeds, RAN, Network Measurement, Experiment Design, Wireless Prototyping, Multi-Band Systems, Performance Evaluation, RAN Optimization

Selected Research Publications

AI-Powered Resilience: A Dual-Approach for Outage Management in Dense Cellular Networks — *Computer Communications* 2025 | DOI

On Multi-Parameter Optimization and Proactive Reliability in 5G and Beyond Cellular Networks — *Sensors 2025* 2025 | DOI

An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin — *IEEE PIMRC* 2024 | DOI

Refining Wireless Propagation Models using Domain-Informed GANs amid Data Scarcity — *IEEE VTC-Fall 2024* 2024 | Link

Academic Service and Recognition

Academic Contributions

- Contributed to course material, examinations, curriculum alignment and student academic guidance.
- Supported laboratory documentation, technical reviews and student-project assessment.

Selected Awards

William H. Barkow Scholarship *2024*
University of Oklahoma, Electrical and Computer Engineering

Clyde L. Farrar Endowment Scholarship *2024*
University of Oklahoma

P.H. Robinson Fellowship *2023*
University of Oklahoma

FAS-Charles Forgan Scholarship *2023*
University of Oklahoma

Relevant Professional Development

1. 4G Network Fundamentals — Mines-Télécom/Coursera | Completed | Aug 2026
LTE radio and core networks, cellular architecture, mobility management, and 4G fundamentals.

2. AI Foundations & Industry Overview for Telecommunication — Coursera | Completed | Jun 2026
AI applications in telecom networks, operations analytics, service assurance, and automation.

3. Network and Security Optimization in Telecommunication — Coursera | Completed | Jun 2026
Data-driven network optimization, telecom security, reliability analysis, and operational efficiency.

4. Python for Data Science, AI & Development — IBM/Coursera | Completed | Jul 2026
Python programming, data analysis, APIs, and reproducible data-science workflows.